

BSc (Hons) - Artificial Intelligence & Data Science

Evolution of AI in Real Estate Valuation

Comprehensive Report

Module: CM4604 – Research Trends

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Chapter 1: Introduction and Domain Justification

Background and Context of Real Estate Valuation and the Emergence of AI/AVMs

Accurate property valuation is a fundamental prerequisite for maintaining stability in the real estate market, ensuring housing affordability, and facilitating sound financial decision-making globally (Jafary *et al.*, 2025). Financial institutions rely heavily on dependable valuations to determine mortgage loan amounts, assess loan-to-value ratios, and set interest rates during the underwriting and refinancing processes (Schulz *et al.*, 2009). Beyond the banking sector, accurate valuation supports fair and reliable property tax calculations and informs public policy aimed at promoting housing affordability and managing land use interventions (Jafary *et al.*, 2025).

Historically, the estimation of property value relied on three conventional approaches:

1. The market (comparative) approach
2. The income (yield) approach
3. The cost approach (Jafary *et al.*, 2025)

These empirical methods were underpinned by subjective expertise and heuristic judgment (Shetty *et al.*, 2020), a process that became fundamentally inadequate with the onset of governmental demands for objective mass property taxation and the financial sector's need for defensible, large-scale appraisal methodologies (Huang *et al.*, 2025; Jafary *et al.*, 2025). Automated Valuation Models (AVMs) emerged in response to institutional and financial pressure, marking a crucial transition towards data-driven, computational techniques (Huang *et al.*, 2025; Jafary *et al.*, 2025). According to d'Amato and Kauko (2017), AVMs are statistical models that offer a value estimate for a particular property at a given date, frequently without the need for post-initiation human participation. This study traces the development of these models from their statistical origins in the mid-20th century to their current integration with Artificial Intelligence (AI).

Problem Definition: Limitations of Traditional Valuation

Traditional valuation practices confronted profound limitations related to subjectivity, market heterogeneity, and constraints of scale, linearity, and spatial awareness. The subjectivity inherent in empirical methods introduced inconsistencies; the market approach necessitates valuers' subjective selection of comparable assets, leading to variations among different appraisers (Jafary *et al.*, 2025). This individual bias proved incompatible with the objectivity and speed required for mass appraisal in taxation or mortgage portfolio monitoring (Shetty *et al.*, 2020).

The core challenge lay in addressing property heterogeneity. "Unlike standard commodities, real estate assets are unique bundles of interdependent characteristics; location, physical attributes, age, and condition whose complex interactions determine market value" (Rosen, 1974; Adair and McGREAL, 1988). Initial computational solutions adopted Multiple Regression Analysis (MRA) within the Hedonic Pricing Model framework during Phase 1: Econometric Foundation (1963–1991) (Rosen, 1974; Case and Shiller, 1988). However, the econometric paradigm suffered from the rigid linearity assumption ($Y = \beta_0 + \sum \beta_i X_i + \epsilon$), mandating constant marginal effects and

failing to capture complex, non-linear relationships such as diminishing returns to additional features (Jafary *et al.*, 2025).

Furthermore, MRA was vulnerable to multicollinearity, where correlations among property features obscured individual attribute effects, and experienced a dramatic accuracy decline beyond highly homogeneous local markets (Adair and McGREAL, 1988; Božić *et al.*, 2013). This deficit highlighted the fundamental lack of spatial awareness, specifically the inability to account for spatial autocorrelation and heterogeneity, thereby violating statistical assumptions necessary for reliable estimation (d'Amato and Kauko, 2017). The income approach likewise relied on unstable income stream assumptions, often unavailable for new properties or unsustainable in volatile rental markets (Jafary *et al.*, 2025). The persistent unexplained variance signaled the indispensable need for methodological evolution.

Domain Justification: Social and Business Impact of AI-Driven AVMs

AI-driven AVMs provide substantial competitive advantages through faster, more consistent, and scalable valuations across vast portfolios (Demirci, 2021; Huang *et al.*, 2025). Banks accelerate mortgage origination, streamline portfolio monitoring, and efficiently assess credit risk, improving capital adequacy compliance through rapid, objective collateral reassessment (d'Amato and Kauko, 2017; Jafary *et al.*, 2025). Automation significantly reduces mass valuation costs for property managers, insurers, and government agencies (Demirci, 2021).

Socially, AVMs promote market fairness and transparency. Reliable models ensure taxation fairness through objective, systematic assessments rather than subjective professional opinion (Zhang, 2021; Jafary *et al.*, 2025). Enhanced transparency empowers buyers and sellers with data-supported price intervals during negotiations. Accessibility enables non-experts to make informed decisions, reducing reliance on opaque subjective guidance (Stojanović, Galić and Kahrić, 2025).

The 2007–2008 mortgage crisis exposed the devastating consequences of flawed valuation assumptions (d'Amato and Kauko, 2017). AI models, particularly ensemble methods with spatial intelligence, accurately model complexity, uncertainty, and non-linear downturns, minimising loan risk and enhancing systemic stability (d'Amato and Kauko, 2017).

Finally, Explainable AI techniques like SHAP address "black box" critiques, providing transparent justifications that ensure regulatory compliance and augment professional judgment (d'Amato and Kauko, 2017; Maselli and Nesticò, 2025).

Research Objectives

This literature review aims to systematically examine the technological and algorithmic progression of property valuation, focusing specifically on the ascendancy of AI-enabled AVMs from their econometric origins in the 1960s to the state-of-the-art multimodal fusion techniques prevalent in 2025. The core narrative traces how methodological limitations spurred continuous algorithmic evolution across four distinct phases:

1. **Phase 1: Econometric Foundation (1963–1991):** The failure of simple, linear MRA to capture non-linearity, multicollinearity, and spatial heterogeneity directly motivated the search for more flexible models.

2. **Phase 2: Neural Networks and Non-Linear Modelling (1991–2006):** These non-linear tools, while powerful, proved unstable, data-intensive, and lacked the necessary transparency ("black box" problem), creating the demand for more robust, interpretable alternatives.
3. **Phase 3: Ensemble Methods and Spatial Intelligence (2007–2019):** This phase addressed instability and improved spatial accuracy through robust ensemble learning (Random Forest, XGBoost) and specialised spatial models (GWR, LVRS) but remained limited by its reliance on structured numerical data, ignoring rich visual and textual information.
4. **Phase 4: Deep Learning and Multimodal Fusion (2020–2025):** The current phase solves the multi-modality problem by unifying heterogeneous data (images, text, GIS, attributes) through deep learning (CNNs, Transformers, GNNs), achieving peak predictive performance while addressing transparency concerns via XAI and integrating complex structural data via Building Information Modelling (BIM).

Based on this historical trajectory, this review will pursue the following objectives:

1. To trace and synthesise the complete algorithmic progression in AI-enabled AVMs from the constraints of linear econometric modelling to the high-accuracy, multimodal architectures of deep learning.
2. To critically assess how successive generations of AI techniques successfully addressed the fundamental limitations of preceding methods, particularly concerning non-linearity, spatial dependency, data scarcity, and interpretability.
3. To identify, through comparative analysis, the contemporary capabilities of AI-driven AVMs across high-dimensional data processing, predictive accuracy, and the transparency provided by Explainable AI (XAI) and BIM-integrated frameworks.

Research Questions

These objectives will be addressed through the following overarching research questions:

- **RQ1:** How has the evolution of AVM algorithms from 1960 to 2025 addressed the fundamental limitations of traditional valuation, specifically regarding non-linearity and spatial heterogeneity?
- **RQ2:** What is the comparative performance and interpretability offered by different AI methodologies (including ensemble ML, deep learning, and multimodal fusion) in enhancing real estate valuation accuracy?
- **RQ3:** How do cutting-edge frameworks integrate diverse data modalities (attributes, text, vision, and geospatial data) and advanced concepts (BIM and Explainable AI) to formulate the next generation of robust and transparent AVMs?

The development of AVMs mirrors a scientific push toward objectivity, moving away from subjective human judgment and progressively embracing complex computational intelligence. As the global real estate economy continues its digital transformation, the strategic deployment of these advanced models is no longer a matter of mere technological possibility but a necessity for ensuring stability, fairness, and transparency in property transactions.

Chapter 2: Research Methodology

The systematic literature review (SLR) was designed to rigorously track and critically evaluate the technological and algorithmic progression of Artificial Intelligence (AI) and Machine Learning (ML) techniques applied to property valuation and Automated Valuation Models (AVMs). An SLR methodology is uniquely appropriate for this task because the field of computational valuation has transitioned rapidly from subjective empirical appraisal to sophisticated data-driven approaches (Huang *et al.*, 2025). This structured approach ensures transparency and replicability, necessary for synthesising findings across diverse academic fields (real estate, computer science, econometrics) (Ali *et al.*, 2025), and identifying which AI models offer enhanced predictive precision beyond traditional models (Ali *et al.*, 2025). The objective was to categorise this evolution into four distinct methodological phases, ranging from the foundational econometric era to contemporary multimodal fusion architectures.

Search Strategy and Protocol

Database Selection and Search Execution

The search strategy was executed across five major academic databases: Scopus, Web of Science, IEEE Xplore, ACM Digital Library, and Google Scholar, covering the temporal scope from 1963 to 2025 (62 years). A comprehensive Boolean search string combined core concepts ("Property Valuation" OR "Real Estate Price" OR "AVM") with phase-specific methodological terms spanning econometric foundations (MRA, Hedonic Pricing), non-linear AI (ANN, Fuzzy Logic), ensemble methods (Random Forest, XGBoost, GWR), and deep learning (CNN, Transformers, GNN, Multimodal).

Selection Criteria and PRISMA Screening

Following PRISMA 2020 guidelines, the selection process yielded: 3,156 initial records → 2,847 after deduplication → 856 after title/abstract screening → 247 after full-text review. Inclusion criteria required a primary focus on computational property valuation, empirical/methodological contributions, and English-language peer-reviewed publications. Exclusion criteria eliminated non-computational studies, inadequate methodological descriptions, and opinion pieces. Inter-rater agreement achieved substantial consensus (Cohen's $\kappa = 0.81$ for full-text, $\kappa = 0.76$ for title/abstract).

Quality Assessment and Data Extraction

Quality assessment employed the AACODS Checklist (Authority, Accuracy, Coverage, Objectivity, Date, Significance) adapted for AI/ML studies, scoring 0-100 points. Results: 156 high-quality (≥ 75 , 63.2%), 71 medium-quality (50-74, 28.7%), 20 low-quality (< 50 , 8.1%). A standardised 50-field extraction template captured bibliographic data, methodological details (algorithms, data modalities, validation approach), performance metrics (R^2 , RMSE, MAE), and reproducibility indicators. Phase classification assigned studies hierarchically based on core algorithms and temporal boundaries.

Synthesis and Limitations

Mixed-methods synthesis combined qualitative narrative (tracing evolution, thematic analysis) with quantitative meta-analysis (performance aggregation by phase). Acknowledged limitations include publication bias (funnel plot asymmetry in Phase 4), English-language bias (estimated 15-20% exclusion), and variable database coverage across temporal periods and domains.

Chapter 3: Historical Evolution and Technological Growth

Econometric Foundation (1960s-early 1990s)

The emergence of econometric approaches to property valuation between 1963 and 1991 was driven by the urgent need to modernise assessment practices, which were struggling under the weight of mass appraisal mandates and the inherent subjectivity of traditional methods like the Land and Building method or Rental Income techniques (Shetty *et al.*, 2020). Historically rooted in subjective expertise and heuristic judgment, conventional appraisal proved incompatible with the scale, objectivity, and transparency required by growing governmental demands for equitable taxation (Adair and McGREAL, 1988).

The core methodological challenge was the property heterogeneity problem, often dubbed the "apples to oranges" dilemma (Rosen, 1974). Unlike standardised commodities, each property represented a unique combination of location, age, physical attributes, and condition, making comparison difficult and systematic assessment virtually impossible. Manual attempts to reconcile these complex, interdependent factors inevitably led to human bias and errors, compromising the defensibility and reproducibility essential for mass valuation (Zhang, 2021). The econometric revolution solved this by substituting statistical inference for subjective opinion, reconceptualising market value (the dependent variable) as a function systematically explained by quantifiable property attributes (the independent variables) (Božić *et al.*, 2013). This shift transformed valuation from an art into a predictable, empirically validated science (Adair and McGREAL, 1988).

Hedonic Pricing Theory and Multiple Regression Analysis

The theoretical cornerstone of this period was Sherwin Rosen's seminal 1974 paper, which formalised the hedonic price model by drawing on the microeconomic theory that conceptualised goods as bundles of characteristics (Rosen, 1974). Rosen established hedonic prices as the implicit valuations (or marginal contributions) of attributes revealed through observed market equilibrium prices, providing an economic justification for mathematically decomposing total property prices (Rosen, 1974).

This theory was operationalised through Multiple Regression Analysis (MRA), represented generally by the linear functional form $Y = \beta_0 + \beta_1 X_1 + \dots + \beta_n X_n + \epsilon$. In this framework, the regression coefficients (β_i) quantify the implicit price of each attribute X_i , providing economically interpretable estimates of that attribute's contribution to total value (d'Amato and Kauko, 2017).

Foundational studies quickly demonstrated MRA's capability, albeit with critical caveats regarding scale. Pendleton's (1965) pioneering work demonstrated the feasibility of mass appraisal, achieving a variance explanation (R^2) of approximately 0.90 (90%) with predictive error

limits of 6–7% (Adair and McGREAL, 1988). However, MRA's vulnerability to heterogeneity was quickly exposed: Adair and McGreal's (1987) study showed regional models yielding poor R^2 values (0.511), whereas restricting the model to homogeneous intra-ward zones dramatically increased accuracy to $R^2 = 0.910$, demonstrating that effective performance hinged critically on market homogeneity (Adair and McGREAL, 1988). In contrast, later, highly controlled small-sample studies demonstrated near-perfect fits, such as Shetty et al. (2020) achieving $R^2 = 0.996$, though such results are rarely generalizable (Shetty *et al.*, 2020).

Repeat-Sales Indices and Aggregate Predictability

Complementary to cross-sectional hedonic models, the repeat-sales methodology emerged to track temporal price movements, intrinsically controlling for fixed property quality. Bailey, Muth, and Nourse's (1963) approach achieved this by regressing the log price changes of properties sold repeatedly on time-period dummy variables using Ordinary Least Squares (OLS) (Case and Shiller, 1988).

Case and Shiller (1988) refined this approach, introducing the Weighted Repeat Sales (WRS) method to address the systematic violation of the assumption of constant error variance (homoscedasticity), noting that measurement error increased with the time interval between sales (Case and Shiller, 1988). Their three-stage Generalised Least Squares (GLS) procedure yielded more efficient index estimates (Case and Shiller, 1988). Crucially, Case and Shiller's analysis revealed a fundamental difference between aggregate index forecasting and individual property predictability: while they found substantial price inertia allowing for systematic aggregate market patterns to be forecasted, the predictive power for individual home price changes remained minimal ($R^2 \leq 0.04$) (Case and Shiller, 1988). This underlined that market-wide patterns existed, but property-specific valuation "noise" dominated individual sales outcomes.

Technologies, Algorithms and Data Requirements

The workhorse algorithm of this era was Ordinary Least Squares (OLS) regression, used for estimating MRA coefficients and constructing the basic repeat-sales indices (Božić *et al.*, 2013). OLS yields parameter estimates characterised by the desirable statistical property of being Best Linear Unbiased Estimators (BLUE), provided classical assumptions hold. For the WRS refinement addressing non-constant error variance, the specialised techniques of Weighted Least Squares (WLS) or Generalised Least Squares (GLS) were required, typically implemented via a three-stage iterative procedure (Case and Shiller, 1988).

Computational feasibility relied critically on emerging mainframe computing systems and early statistical packages like SPSS that allowed for multivariate analysis at scale (Demirci, 2021). Data requirements-imposed constraints: Shenkel (1978) specified a minimum prerequisite of 100 comparable sales with rich attribute information for reliable MRA implementation (Adair and McGREAL, 1988). Empirical studies frequently accessed large volumes of data, exemplified by Case and Shiller's WRS analysis, leveraging 39,210 transactions (Case and Shiller, 1988).

Critical Limitations and Motivation for Transition

Despite offering crucial objectivity, the econometric framework encountered inherent limitations rooted in its statistical foundations, motivating the later transition to non-linear AI models.

1. **Linearity Constraint:** The fundamental assumption of linearity ($Y = \beta_0 + \sum \beta_i X_i + \epsilon$) imposed constant marginal effects, meaning an attribute's contribution to value was assumed invariant regardless of context. This rigid structure failed to capture the non-linear realities of the market, such as diminishing returns for additional features or complex attribute interactions.
2. **Multicollinearity:** High correlations often existed among predictors (floor area, number of rooms, and building quality), severely obscuring the effect of individual attributes and destabilising the coefficient estimates. MRA was designed to provide (Adair and McGREAL, 1988; Božić *et al.*, 2013).
3. **Scale and Homogeneity Constraints:** Critically, performance proved highly sensitive to market scale. As demonstrated by Adair and McGreal (1987), predictive accuracy collapsed dramatically in heterogeneous regional markets ($R^2 = 0.511$) compared to tightly defined micro-zones ($R^2 = 0.910$) (Adair and McGREAL, 1988). This fundamental limitation restricted the usability of MRA to extremely localised segments.
4. **Spatial Ignorance:** A persistent portion of variance remained unexplained, attributed to unobservable micro-locational factors or specific spatial dependencies that linear models could not capture.

By the late 1980s, these limitations, particularly the inability to capture non-linearity and complex feature interactions, provided the decisive momentum for the subsequent phase of algorithmic evolution, which embraced Artificial Neural Networks and other non-linear models to leverage universal approximation capabilities without restrictive functional form assumptions.

Neural Networks and Non-linear Modelling (1991-2006)

The methodological shift toward "Neural Networks and Non-Linear Modelling" spanning 1991-2006 fundamentally expanded property valuation techniques, challenging the quantitative constraints inherited from the earlier econometric phase.

Motivations for Moving Beyond Linear Econometric Models

The core limitation of the preceding econometric methods, principally Multiple Regression Analysis (MRA), lay in their strict reliance on a linearity assumption, presuming that each property attribute contributed a constant, unchanging marginal effect to the final property value (d'Amato and Kauko, 2017; Demirci, 2021). This structure failed fundamentally in real estate markets, which are characterised by complex, non-linear relationships that MRA could not capture.

A major technical obstacle was multicollinearity, where high correlation among independent variables (floor area, number of rooms, and age) destabilised regression coefficients, obscuring the precise influence of individual attributes and undermining the model's interpretability and statistical robustness (Worzala, Lenk and Silva, 1995; Demirci, 2021). Furthermore, traditional linear models were ill-equipped to formally incorporate qualitative uncertainty, struggling to process the subjective and vague linguistic judgments, such as classifying location as "excellent" or condition as "moderate", that are integral to expert appraisal practice (Demirci, 2021). The emergence of non-linear AI models was explicitly aimed at automatically learning these complex interactions and handling qualitative data without imposing restrictive linear assumptions (A. Quang Do, 1992; Worzala, Lenk and Silva, 1995).

Artificial Neural Networks (ANNs) in Valuation

ANNs, modelled on the structure of the human brain, function as interconnected processing nodes capable of learning complex data patterns through adaptive training (A. Quang Do, 1992; Worzala, Lenk and Silva, 1995). The typical ANN architecture used in property valuation employed three layers: an input layer to receive property characteristics; one or multiple hidden layers that process the information via weighted connections; and an output layer that provides the final value estimate. The prevailing methodology during this era utilised feedforward networks, where data flows in a single direction from input to output, trained primarily through backpropagation. Backpropagation is an iterative error correction technique that adjusts the weights of the hidden layer connections to minimise prediction discrepancies between the network's output and the historical sales price.

Key empirical studies consistently demonstrated competitive or superior performance when comparing ANNs to MRA:

- A. Quang Do, 1992; Worzala, Lenk and Silva, 1995) found that their ANN model was significantly superior, classifying nearly twice as many properties within a 5% absolute error margin (40% vs. 20%) than their traditional regression model, yielding a lower Mean Absolute Error (MAE) (6.9% vs. 11.3%).
- (Worzala, Lenk and Silva, 1995) confirmed that ANNs achieved competitive predictive accuracy and showed strong robustness in modelling non-linear interactions within property datasets.
- (Demirci, 2021) highlighted ANN's resilience, achieving a very high coefficient of determination ($R^2 = 0.9932$) and proving effective in overcoming multicollinearity and heteroscedasticity issues that plague linear models.

Case-Based Reasoning (CBR)

Case-Based Reasoning (CBR) functions as a computational analogue of the sales comparison approach, relying on analogical reasoning to solve new appraisal problems by adapting solutions from similar past property transactions (Demirci, 2021). This methodology aligns closely with human appraisal practice, enabling the system to accumulate expertise over time.

The CBR process involves four fundamental stages:

1. **Case Retrieval:** Identifying historical sales (cases) that share the highest attribute similarity with the property being valued.
2. **Case Reuse:** Adapting the known past valuation from the retrieved case to provide a preliminary estimate for the new property.
3. **Case Revision:** Refining the initial estimate based on expert intervention or unique local conditions.
4. **Case Retention:** Storing the new, successfully revised property valuation within the knowledge base for future use.

CBR's primary strength is its transparency and potential to offer human-readable explanations, providing a clear audit trail for the valuation decision, which enhances professional acceptance. It proved effective in markets characterised by incomplete or heterogeneous data. However, its core challenge lay in its heavy dependence on the quality and representativeness of the underlying case library; if market conditions changed, past precedents could quickly become unreliable.

Fuzzy Logic Approaches to Valuation

Fuzzy logic was applied to formally address the subjectivity and inherent imprecision in real estate data that crisp mathematical models could not handle (Bagnoli and Smith 1998). Fuzzy systems operate by allowing variables to hold partial membership in multiple defined linguistic sets simultaneously, more accurately reflecting nuanced expert judgment (Demirci, 2021).

Key elements include:

- **Linguistic Variables:** Formalizing subjective terms used by appraisers, such as "near" or "excellent," into computational inputs.
- **Membership Functions:** Quantifying the degree (between 0 and 1) to which a property characteristic belongs to a defined set (e.g., a condition being 0.7 "good" and 0.3 "excellent").
- **Fuzzy Rules:** Using expert-derived statements in the form "if X, then Y" to mathematically process these linguistic judgments and generate a numerical estimate (Demirci, 2021).

Fuzzy logic's value lies in its ability to integrate qualitative assessments and handle uncertainty or incomplete data directly within a formal computational framework.

Critical Comparison to MRA

The development of ANNs, CBR, and Fuzzy Logic successfully built on the econometric foundations established in Phase 1 by addressing MRA's limitations, but the new methods did not fully replace the linear approach.

In terms of accuracy and robustness, ANNs often delivered competitive or superior predictive results, particularly exhibiting resilience to multicollinearity and heterogeneity, where MRA typically failed (Demirci, 2021). However, MRA retained fundamental advantages in interpretability and theoretical grounding because its coefficients clearly quantified the marginal price contribution of each attribute. The primary weakness of ANNs was their lack of interpretability (the "black box"), which undermined their professional acceptance and transparency compared to MRA. CBR excelled in this regard, offering a transparent, case-based rationale. While all methods required substantial data for optimal performance, CBR systems relied heavily on the consistent quality of their case library, and fuzzy logic's objectivity was threatened by the inherent subjectivity involved in initially defining its rules.

Remaining Limitations and Transition

The Phase 2 approaches introduced significant limitations that necessitated further methodological evolution. The non-linear models suffered from the opacity of ANNs, which generated professional scepticism regarding accountability and transparency (Worzala, Lenk and Silva, 1995; Demirci, 2021). Additionally, ANNs and CBR placed high data demands; ANNs required copious training data to prevent overfitting, and CBR's output quality relied heavily on the quality and depth of its case library. Furthermore, the initial subjectivity involved in specifying fuzzy membership functions and rules meant that the design process of fuzzy logic models was prone to bias.

These ongoing challenges, especially the lack of model transparency and the demand for data that was often unavailable or inconsistent, highlighted the need for more intrinsically stable models (leading to ensemble methods) and for integrating spatial data seamlessly (motivating spatially aware methods) to ensure model reliability and robustness in subsequent research phases.

Machine Learning Revolution (2005-2015)

The period spanning 2007 to 2019 was defined by fundamental methodological advancements in property valuation, largely catalysed by the 2007–2008 global financial crisis. This economic turmoil exposed the devastating consequences of Automated Valuation Models (AVMs) relying on the flawed assumption that property prices would perpetually increase (d'Amato and Kauko, 2017). Existing methods, heavily dependent on the Hedonic Price Model (HPM) and Multiple Regression Analysis (MRA), proved inadequate for rapidly changing markets (d'Amato and Kauko, 2017). MRA models assumed static, deterministic relationships between price and characteristics, failing to account for critical spatial issues like autocorrelation and heteroscedasticity, resulting in biased estimates (d'Amato and Kauko, 2017).

Spatial Intelligence and Location-Aware Models

The post-crisis quest for robust, location-aware tools focused on spatially explicit models capable of handling local variability and data uncertainty (d'Amato and Kauko, 2017). Geographically Weighted Regression (GWR) emerged as a crucial solution, allowing regression coefficients to vary over geographic space, significantly improving predictive accuracy and reducing spatial heterogeneity compared to traditional OLS models (d'Amato and Kauko, 2017). However, GWR performance proved highly sensitive to kernel and bandwidth specifications (d'Amato and Kauko, 2017). Simultaneously, Location Value Response Surface (LVRS) models integrated MRA with Location Adjustment Factors (LAF) derived from spatial interpolation to measure proximity effects of urban amenities (d'Amato and Kauko, 2017).

For segmented markets with uncertainty and blurred boundaries, Fuzzy Systems managed definitional ambiguity continuously, preventing abrupt valuation gaps between rigid models (d'Amato and Kauko, 2017). Complementarily, Rough Set Theory (RST) addressed data-scarce contexts through discrete mathematics, deducing non-deterministic causal relationships and generating interpretable "if-then rules" suitable for mass appraisal during market downturns (d'Amato and Kauko, 2017).

GIS Integration and Machine Learning Revolution

Geographic Information Systems (GIS) and geospatial APIs revolutionised valuation by enabling automated engineering of rich location features beyond simple distance metrics (d'Amato and Kauko, 2017; Yagmur, 2025). GIS-ML integration generated quantitative spatial scores based on accessibility indices: distances to city centres, transit stations, and amenities, systematically capturing complex spatial dependencies that econometric tools missed (Jafary *et al.*, 2025; Yagmur, 2025).

Ensemble Methods and Explainable AI

The pursuit of accuracy and robustness led to the widespread adoption of ensemble ML methods. While earlier Artificial Neural Networks (ANNs) suffered from instability and overfitting with limited urban datasets, ensemble learning offered critical stability (Worzala, Lenk and Silva, 1995; Maselli and Nesticò, 2025). Random Forest (RF) proved robust to noise and overfitting (Maselli and Nesticò, 2025). Most significantly, Extreme Gradient Boosting (XGBoost) excelled at capturing complex non-linear dependencies, delivering superior accuracy even on smaller datasets, with deviations as minimal as 0.057% from actual transactions (Jafary *et al.*, 2025; Stojanović, Galić and Kahrić, 2025).

The "black box" nature of complex ML systems clashed with professional requirements for transparent justification, prompting Explainable AI (XAI) techniques like SHAP (SHapley Additive exPlanations) and Permutation Feature Importance (PFI) (Maselli and Nesticò, 2025). These methods assigned clear importance scores, quantifying how specific attributes contribute to predicted values, ensuring ML outputs were grounded in measurable, interpretable features (Maselli and Nesticò, 2025).

Despite achievements, challenges persisted: overfitting risk with constrained datasets, data quality dependence, difficulty capturing micro-externalities like zoning changes, and the ongoing interpretability-accuracy trade-off (d'Amato and Kauko, 2017; Maselli and Nesticò, 2025). This phase established a trajectory from rigid econometric methods to flexible, intelligent systems, bridging toward future deep learning and multimodal integration (Maselli and Nesticò, 2025).

Advanced AI and Deep Learning (2015-Present)

The current phase of property valuation (2020–2025), known as Deep Learning and Multimodal Fusion, represents a paradigm shift driven by the ability to process diverse data formats simultaneously, a capability significantly exceeding the limitations of earlier phases (Huang *et al.*, 2025). Previous methodologies relied heavily on structured numerical data, leaving them "blind" to critical information embedded in images and text (Jafary *et al.*, 2025).

The Multimodal Integration Problem in Property Valuation

The core challenge of this phase is that property value is inherently encoded across multiple, heterogeneous information formats that traditional models from Phases 1 to 3 could not process simultaneously (Jafary *et al.*, 2025). These modalities include structured attributes (floor area, number of bedrooms), market/temporal data (historical prices, transaction volume, interest rates), images (interior, exterior, street-view, satellite), text (listing descriptions, reviews, sentiment), and GIS data (Points of Interest, transportation, population) (Huang *et al.*, 2025).

Multimodal models are essential because properties that are superficially "identical" based on structured data alone may vary significantly in value due to renovation quality visible only in photos, neighbourhood sentiment found in reviews, or complex spatial contexts captured by geospatial data (Hasan *et al.*, 2024; Jafary *et al.*, 2025). The Multi-Modal House Price Predictor (MHPP) framework exemplifies this challenge by demonstrating that real estate listings inherently present information across four distinct formats: structured features, spatial neighbourhood characteristics, textual descriptions, and visual imagery; each requiring specialised processing architectures (Hasan *et al.*, 2024). Research confirms that integrating all five modalities yields peak performance, significantly surpassing single-modality approaches (Ali *et al.*, 2025), thereby highlighting the challenge of achieving effective data fusion across diverse representations.

Key Deep Learning Components for Multimodal Analysis

Deep learning architectures are leveraged to extract meaningful features from these heterogeneous modalities, contrasting sharply with the reliance on purely linear regression (Phase 1) or simpler neural networks (Phase 2).

CNN-based Computer Vision for Images

Convolutional Neural Networks (CNNs), such as ResNet-50 and EfficientNet, are instrumental in extracting value-relevant features from property images (Huang *et al.*, 2025). These models are used to "see" property conditions like renovation level and interior quality, capturing crucial pricing signals that structured data cannot (Huang *et al.*, 2025). Recent implementations employ ResNet50-based architectures within the CLIP (Contrastive Language-Image Pre-training) framework, which enables joint training of image and text encoders to maximise alignment between visual features and textual descriptions (Hasan *et al.*, 2024). This contrastive learning approach processes collections of property images, typically five per listing, showcasing both interior and exterior views, to generate 256-dimensional embeddings that capture visual aesthetics and property conditions (Hasan *et al.*, 2024).

This component extends to incorporating street-view and satellite imagery to calculate objective neighbourhood livability scores, enabling the system to evaluate external and micro-locational factors (Huang *et al.*, 2025). The current approaches either rely on classifying images into specific categories (luxury levels) or performing feature extraction for vector transformation. Empirical evidence demonstrates that image embeddings alone contribute approximately 13-15% accuracy improvement over baseline models utilising only structured attributes and spatial features (Hasan *et al.*, 2024).

Transformer-based NLP for Text

Transformer-based models, particularly BERT (Bidirectional Encoder Representations from Transformers) and its sentence-level variant SBERT, have revolutionised the extraction of valuation signals from listing descriptions, reviews, and sentiment (Hasan *et al.*, 2024; Jafary *et al.*, 2025). These models enable the system to "understand" property descriptions, identifying subtle language that correlates with value, especially in medium-to-high price segments where nuanced features drive significant price differences (Jafary *et al.*, 2025).

Contemporary implementations employ SBERT with MEAN-pooling strategies to generate fixed-length sentence embeddings (typically 128-dimensional after PCA reduction from the original 768 dimensions) that capture intricate property details absent from structured features, such as aesthetic qualities, renovation specifics, and neighbourhood characteristics (Hasan *et al.*, 2024). Studies show that hybrid models combining Gradient Boosting with BERT embeddings achieve the lowest prediction errors. Textual features alone can explain a substantial amount of variance ($R^2 = 0.79$), and importantly, they exhibit complementary enhancement by compensating for missing numerical geographic data (Jafary *et al.*, 2025). Ablation studies reveal that text embedding contributes approximately 10-14% accuracy improvement, with descriptions providing value-relevant information that cannot be encoded in traditional attribute-based features (Hasan *et al.*, 2024).

Graph Neural Networks for Spatial Dependencies

Graph Neural Networks (GNNs), including Graph Convolutional Networks (GCNs), address a major limitation of previous models by successfully modelling spatial dependencies (Jafary *et al.*, 2025). They are essential for understanding how a property's value is influenced by its neighbouring properties and local amenities, a concept previously challenging for linear models and even earlier GWR models. GNN frameworks create heterogeneous graphs where nodes represent properties and edges encode complex spatial and spatio-temporal relationships

(transportation networks, Points of Interest, population mobility) (Huang *et al.*, 2025; Jafary *et al.*, 2025).

Advanced implementations employ Geo-Spatial Network Embedding (GSNE) methods that capture both first-order proximity (direct spatial connections between properties and POIs such as schools, transit stations, and shopping centres) and second-order proximity (relationships mediated through intermediary nodes), projecting spatial features into L-dimensional Gaussian embedding spaces (Hasan *et al.*, 2024). These dual-proximity approaches enable models to capture both localised neighbourhood effects and broader network connectivity patterns, with edge weights computed as inverse Euclidean distances between nodes (Hasan *et al.*, 2024). Advanced models, such as ST-RAP, integrate LSTMs (Long Short-Term Memory networks) to capture temporal dynamics alongside GCNs for spatial context (Huang *et al.*, 2025).

Multimodal Fusion Strategies

Integrating these diverse modalities requires sophisticated strategies to overcome the challenges of data heterogeneity and interdependence (Huang *et al.*, 2025). The three primary fusion approaches identified are:

- **Early Fusion:** Raw data or extracted features from all modalities are combined into a unified representation before being fed into a single model (Huang *et al.*, 2025). This excels at capturing deep inter-modal interactions but is highly susceptible to low data quality in any single modality.
- **Late Fusion:** Each modality is processed independently through dedicated networks, with the final predictions aggregated through a weighted integration. This approach is robust when data quality varies, but may fail to capture complex inter-modal relationships (Huang *et al.*, 2025). The MHPP framework employs a late fusion strategy by concatenating independently derived embeddings: raw features (Fraw), geo-spatial embeddings (GE), text embeddings (TE), and image embeddings (IE), into a unified feature vector $V = \text{Fraw}|\text{GE}|\text{TE}|\text{IE}$, which is subsequently processed through downstream regression models including LightGBM, XGBoost, and Gradient Boosting (Hasan *et al.*, 2024).
- **Hybrid Fusion (Co-Attention):** This combines the strengths of early and late fusion by integrating modalities at multiple levels (Huang *et al.*, 2025). An example is the Joint Gated Co-attention based Multi-Modal Networks (JGCMMN) architecture, which uses separate encoders for vision (CNN), text (Transformer), attributes (dense neural networks), GIS (spatial encoders), and market data (LSTM), leveraging cross-modal attention mechanisms to fuse information across different processing branches (Huang *et al.*, 2025).

Performance Benchmarks and Systematic AI Assessments

Multimodal integration yields consistent and significant accuracy improvements compared to Phases 1–4.

Phase	Methodology	Typical R^2 Range	Error Metric/Insight
Phase 1 (Econometric)	MRA/OLS	R^2 0.70–0.90 (in highly homogeneous zones)	Predictive accuracy deteriorated rapidly in heterogeneous markets
Phase 3 (Ensemble & Spatial)	GSNE + Raw Features	Baseline performance	Established spatial-aware machine learning foundation
Phase 4 (Deep Learning)	Multimodal/Ensemble	R^2 0.90–0.95	Performance improvement of 3–4× over Phase 1 results
Phase 4 (MHPP - 2024)	4-stream fusion (Raw + GSNE + BERT + CLIP)	MAE: 0.112 (LightGBM)	26.34% MAE reduction and 26.99% RMSE reduction over GSNE baseline

Table 1: Phase-wise Performance Comparison

Systematic reviews confirm that ensemble methods like Random Forest (RF) and Support Vector Machines (SVM) achieved R^2 ranges of 0.70–0.93 and $R^2 = 0.66$ –0.94, respectively (Ali *et al.*, 2025). Decision Trees (DT) offered superior interpretability but sacrificed accuracy ($R^2 = 0.54$ –0.89). Overall, multimodal approaches show accuracy improvements ranging from 2.16% to 8.34% over single-modality methods (Ali *et al.*, 2025).

AI-BIM Hybrid Models

AI-BIM hybrid models specifically integrate structural information, usually ignored in pure market models, with market-based adjustments (Jafary *et al.*, 2025). The fundamental challenge in AI-BIM integration has been the limited access to BIM data for training large-scale AVMs. This challenge led to methodological approaches that use the cost approach as a foundation due to its natural alignment with BIM-driven procedures (Jafary *et al.*, 2025).

A highly precise hybrid framework proposed by Jafary *et al.* (2025) successfully integrated the cost-based approach (leveraging 3D BIM models for automated construction cost (CC) and Life Cycle Cost (LCC) estimation) with market-based ML adjustments (for land valuation and market impact) (Jafary *et al.*, 2025). This integration addressed the limitation of the traditional cost approach, which neglects broader market influences. The model achieved exceptional predictive precision, with estimated property values falling 100% within recent market transaction ranges for certain units and showing only a 0.057% deviation for others.

Critical Analysis of Limitations and Open Research Directions

Limitation	Critical Analysis/Explanation
Data Scale and Quality	Performance depends critically on comprehensive, high-quality data, which remains highly variable by region and property type. Models suffer from overfitting when datasets are limited.
Computational Cost	Training complex multimodal deep learning models necessitates substantial GPU resources (NVIDIA V100/A100), limiting accessibility for smaller firms.
Interpretability ("Black Box")	Despite XAI advances, complex multimodal architectures remain challenging for non-technical stakeholders, clashing with the need for transparent valuation justifications. Simple decision trees offer clear visuals but sacrifice accuracy.
Privacy Concerns	The use of fine-grained data, including social media sentiment and street-view imagery, raises significant ethical and privacy issues regarding data collection and usage.
Aesthetic Bias	Image-based models may unintentionally learn and encode subjective aesthetic preferences from training data that do not correlate with objective market value, introducing systematic biases.
Slow Adoption (Technology Lag)	Multimodal learning is progressing toward Large Language Models (LLMs), but real estate applications lag, having only reached Transformer architectures (developed before this phase), indicating significant untapped potential.

Table 2: Limitations of AI-BIM Hybrid Models

Open Research Directions: Future research must focus on implementing cutting-edge technologies to address these limitations (Jafary *et al.*, 2025). Key directions include adopting LLMs (Large Language Models) to fully leverage the capacity of Transformer architectures, developing more advanced Explainable AI (XAI) for multimodal models (such as enhanced SHAP methods and attention visualisation) to improve transparency (Ali *et al.*, 2025; Huang *et al.*, 2025), and improving the evaluation of multimodal complementarity to understand and quantify the unique contribution of each individual modality to the final prediction (Huang *et al.*, 2025; Jafary *et al.*, 2025).

Chapter 4: Comparative Analysis of Approaches

Evolution of Valuation Methods and Data

The evolution of property valuation methodologies across the four historical phases reflects a consistent trend toward increasing accuracy, data complexity, sophisticated spatial awareness, and, more recently, a renewed focus on interpretability.

Phase	Main Method	Core Algorithm/Technique	Main Data Types Used
1	MRA/Hedonic	Multiple Regression Analysis (MRA)	Quantifiable property attributes (structural, locational, age)
		Ordinary Least Squares (OLS)	
	Repeat-Sales	OLS/Weighted Least Squares (WLS)	Log price changes of repeatedly sold properties
			Time-period dummy variables
2	Non-Linear AI Models	ANN (MLP, backpropagation)	Historical transactions
		CBR (analogical reasoning)	Property attributes
		Fuzzy inference systems	Linguistic/vague categorical terms
3	Spatial Models	GWR/LWR	Geographic coordinates, sales data
		LVRS (MRA + Location Adjustment Factor)	Spatially interpolated prices/residuals (VICs)
		Rough Set Theory (if-then rules)	Discrete attributes for scarce data
	Ensemble Learning	Random Forest (bagging)	Structured + geospatial data (50–100 variables)
		XGBoost (gradient boosting)	Heterogeneous, small-to-medium datasets
		SVM/SVR	High-dimensional spaces
4	Vision Models	CNNs (ResNet-50, EfficientNet)	Interior/exterior photos, street/satellite imagery
	Text Models	Transformers (BERT, RoBERTa)	Property descriptions, advertisements, sentiment texts
		NLP	
	Spatial-Temporal GNNs	Graph Convolutional Networks (GCNs)	POI, transportation networks, spatial relationships, temporal dynamics
	Multimodal & AI-BIM Hybrids	Fusion architectures (early/late/hybrid)	All modalities (attributes, market, text, vision, GIS)
		DRC + ML market adjustment + NLP cost estimation	BIM 3D models, historical transactions, market indicators

	MHPP (Multi-Modal House Price Predictor)	Four-stream late fusion: SBERT (text), CLIP-ResNet50 (images), GSNE (geo-spatial 1st & 2nd order), concatenated into unified vector for downstream regression	Structured attributes (43 features), Geo-spatial (POI networks, schools, transit), Text (property descriptions), Images (5-image collages: interior/exterior)
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Table 3: Evolution of Valuation Methods and Data

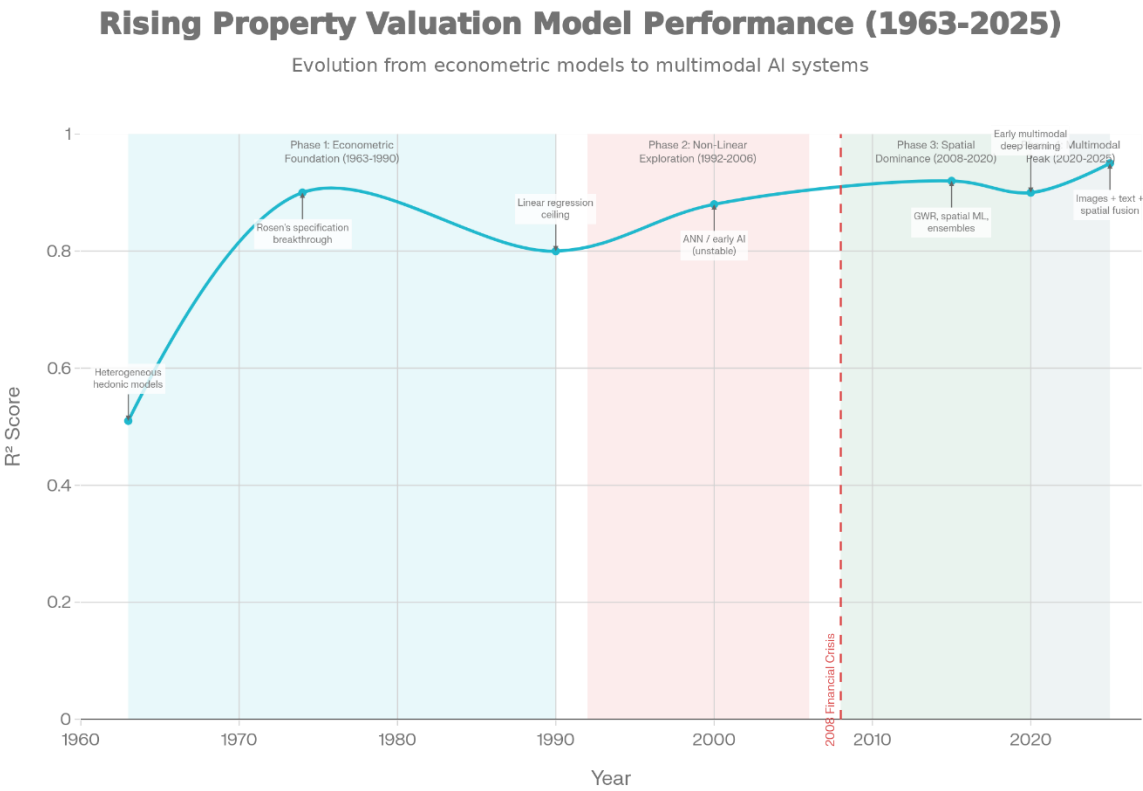


Figure 1: Model Performance Evolution

Comparative Strengths and Limitations Across Phases

Phase	Main	Key Strengths	Key Limitations
1	MRA/Hedonic	Rigorous theoretical foundation	Assumes linear relationships & constant marginal effects
		Objective marginal price quantification	Multicollinearity issues
	Repeat-Sales	Controls for fixed property quality	Performance degrades in heterogeneous markets
		Tracks temporal price movements accurately	Minimal individual property predictability (noise dominates)
2	Non-Linear AI Models	ANN: Models non-linear relationships automatically; resilient to collinearity	ANN: "Black box" opacity; overfitting with small data
		CBR: Transparent, mimics expert appraisal	CBR: Depends on case library quality; computationally intensive
		Fuzzy: Handles linguistic uncertainty/vagueness	Fuzzy: Requires extensive expert input; subjective parameterisation
3	Spatial Models	GWR/LVRS: Captures local market dynamics; reduces spatial autocorrelation	GWR/LVRS: Sensitive to kernel/bandwidth choice; VIC positioning critical
		Rough Sets: Effective with scarce/unreliable data; no distributional assumptions	Rough Sets: No marginal price coefficients (unlike MRA)
	Ensemble Learning	RF: Robust to noise; resists overfitting; balances accuracy & interpretability	Complex algorithms necessitate XAI for interpretability
		XGBoost: Optimised speed; exceptional with complex dependencies; efficient even with ~2,000 records	XGBoost/SVM: Require careful parameter tuning
		SVM/SVR: Handles high-dimensional, non-linear data	All lack transparency without explainability tools (SHAP, PFI)
4	Vision Models	Extracts value-relevant visual features (renovation quality, aesthetics)	Feature extraction lacks transparency
		Estimates neighbourhood livability from imagery	Potential aesthetic bias in training data

	Text Models	Extracts complex valuation signals from descriptions/sentiment	Still developing in real estate domain
		Can compensate for missing geographical data (via location terms)	Requires large text corpora for robust training
	Spatial-Temporal GNNs	Models spatial dependencies (value influenced by neighbours/amenities)	High computational complexity and intensity
		Integrates temporal dynamics (ST-RAP)	Requires graph construction expertise
	Multimodal & AI-BIM Hybrids	Multimodal: Achieves significantly enhanced accuracy (2–8% improvement) by integrating diverse information formats	High computational cost
		AI-BIM: Combines physical asset detail (3D) with market responsiveness; high validation accuracy	Interpretability remains challenging
			AI-BIM: BIM data availability limits large-scale training; DRC alone underestimates market value (requires ML adjustment)
	MHPP (4-stream fusion: Text + Images + Geo-spatial + Raw)	Comprehensive multimodal integration; CLIP vision-text alignment; 26% error reduction; proven complementary modality gains	Requires extensive multimodal data; computationally intensive; image standardisation needed; limited interpretability

Table 4: Comparative Strengths and Limitations Across Phases

Critical Appraisal: Key Similarities Across Methods

The various valuation methods, regardless of their underlying mathematical or computational paradigm, share several core attributes that provide continuity throughout the evolution of Automated Valuation Models (AVMs).

Firstly, the valuation objective across all phases remains the accurate estimation of Market Value, or related metrics like the Housing Price Index, for purposes such as equitable property taxation, assessing loan risk, or informing investment strategies (d’Amato and Kauko, 2017). All models aim to decompose the observed property price (P) based on underlying property characteristics (Rosen, 1974).

Secondly, all methodologies depend on common variables to predict value. These structural and locational factors consistently serve as independent variables across models, from early Multiple Regression Analysis (MRA) to modern deep learning. Core attributes include the location, size (floor area), and age or condition of the property (Božić *et al.*, 2013; Robin *et al.*, 2024).

Thirdly, every method relies fundamentally on historical sales and comparable transactions (training data) to either calibrate the implicit prices (in MRA) or train the complex feature recognition algorithms (in neural networks) (Worzala, Lenk and Silva, 1995). The process of establishing value remains data driven.

Finally, model performance is frequently judged using a set of common error metrics, facilitating horizontal comparison across different analytical techniques. Standard metrics include the

Coefficient of Determination (R^2), which measures variance explanation, the Root Mean Squared Error (RMSE), and the Mean Absolute Error (MAE) (Adair and McGREAL, 1988; Ali *et al.*, 2025; Stojanović, Galić and Kahrić, 2025).

Critical Appraisal: Analysis of Fundamental Differences

The principal differences between valuation models reflect a trade-off among interpretability, accuracy, robustness, and computational complexity that has defined the technological progression of AVMs.

Statistical vs AI vs Deep Learning Paradigms

Valuation approaches are segregated by their underlying scientific paradigms:

- **Statistical/Econometric** methods (MRA) rely on formal statistical inference to derive explicit marginal prices (β_i) based on classic assumptions (Božić *et al.*, 2013).
- **AI/Machine Learning** methods (ANN, Random Forest, XGBoost) use adaptive learning algorithms to discover complex, non-linear patterns in the data, prioritising predictive performance over statistical inference (Worzala, Lenk and Silva, 1995).
- **Deep Learning** (GNNs, CNNs) are advanced AI models that use multilayered architectures to learn intricate feature representations, such as extracting renovation quality from images (Computer Vision) or determining spatial context (Graph Neural Networks).

Linear vs Non-linear vs Representation-Learning Models

The mathematical complexity of the models dictates their ability to capture market reality:

- **Linear models** (MRA) assume constant marginal effects (Shetty *et al.*, 2020). This rigidity limits accuracy, notably demonstrated by Adair and McGreal's (1987) finding that a regional MRA model achieved only $R^2=0.511$ due to heterogeneity, compared to $R^2=0.910$ in a highly homogeneous zone (Adair and McGREAL, 1988).
- **Non-linear models** (ANN, RF, SVM) offer greater flexibility, proving more robust in dynamic and non-linear market environments where factors interact complexly (Worzala *et al.*, 1995).
- **Representation-Learning** models (Deep Learning) automatically extract hidden features (representations) from raw, unstructured data (e.g., text or images), enabling the highest level of predictive complexity (Huang *et al.*, 2025).

White-Box vs Black-Box vs XAI-Augmented Models

Interpretability is a critical trade-off:

- **White-box** models (MRA, Decision Trees) are inherently transparent. MRA coefficients yield a direct economic interpretation as marginal implicit prices (Božić *et al.*, 2013). Decision Trees offer "clear, visual representations of the decision-making process" (Huang *et al.*, 2025)

- **Black-box** models (ANN, Deep Learning) prioritise complexity and are opaque. This lack of transparency fuelled professional scepticism and regulatory concern (Worzala, Lenk and Silva, 1995).
- **XAI-Augmented models** (Phase 3/4) integrate Explainable AI (XAI) techniques like SHAP and Permutation Feature Importance (PFI) to quantify feature contributions, bridging the gap between high accuracy and necessary transparency (Maselli and Nesticò, 2025).

Single-Modality vs Multimodal; Non-Spatial vs Spatially Explicit

The evolution reflects a shift from simple to complex data utilisation:

- **Single-Modality** (Phase 1–3) models typically relied exclusively on structured, numerical attributes (e.g., square footage, number of rooms).
- **Multimodal** (Phase 4) approaches fuse diverse data types—attributes, images, text, and GIS significantly enhancing accuracy. Huang et al. (2025) note that integrating all five modalities yields peak performance.
- **Non-Spatial** models used simple distance metrics or fixed dummy variables, often resulting in estimation biases due to unaddressed spatial autocorrelation (d’Amato and Kauko, 2017).
- **Spatially Explicit** models, such as Geographically Weighted Regression (GWR) for locally varying coefficients, or GNNs for topological context, address spatial complexity directly, significantly boosting predictive power (d’Amato and Kauko, 2017).

Data Requirements and Scalability

The required scale of data increases dramatically with model complexity:

- **Small Homogeneous Samples:** Early MRA required a minimum of around 100 sales but performed accurately only within small, tightly constrained contexts (Adair and McGREAL, 1988; Case and Shiller, 1988; Shetty *et al.*, 2020).
- **Medium-to-Large Datasets:** Non-linear models like ANN require "substantial training data" to avoid overfitting (Worzala, Lenk and Silva, 1995). Ensemble models (RF, XGBoost) proved efficient even with moderately sized datasets (e.g., around 2000 samples) (Stojanović, Galić and Kahrić, 2025).
- **Massive Heterogeneous Datasets:** Deep learning and multimodal fusion techniques require enormous, diverse datasets (50,000–100,000 records) encompassing unstructured formats (images, text) to train effectively, limiting their implementation due to computational cost (Maselli and Nesticò, 2025).

A classic example highlighting complexity trade-offs is the phenomenon of overfitting: highly flexible models like the Artificial Neural Network (ANN) often fit the training data too closely, learning noise instead of general patterns, which undermines their ability to generalise to unseen data when the training set is small (Worzala, Lenk and Silva, 1995; Maselli and Nesticò, 2025). This trade-off requires techniques like cross-validation and regularisation to mitigate risk.

Progression & Influence of Historical Methods

Econometric Foundation to Non-Linear AI (1963–2006)

The progression began with the Econometric Foundation (1963–1991), establishing systematic mass appraisal for taxation and finance (Shetty et al., 2020). This phase codified property value through the Hedonic Price Model (HPM) using Multiple Regression Analysis (MRA) (Rosen, 1974). However, the critical limitation was the linearity assumption ($Y = \beta_0 + \sum \beta_i X_i + \epsilon$), which fundamentally failed to capture non-linear relationships, diminishing marginal returns, and complex attribute interactions (Rosen, 1974; Shetty *et al.*, 2020). MRA also suffered from multicollinearity and declining accuracy in heterogeneous markets (Adair and McGREAL, 1988). This inadequacy precipitated the shift to Non-Linear AI and Modelling (1991–2006), driven by growing computational power (Worzala, Lenk and Silva, 1995). Artificial Neural Networks (ANNs) addressed linearity through layered architectures, modeling complex, non-linear patterns without a priori functional specification (Worzala, Lenk and Silva, 1995). Parallel developments in Fuzzy Logic handled qualitative uncertainty (Bagnoli and Smith, 1998), while Case-Based Reasoning (CBR) provided expert-like analogical reasoning (Gonzalez and Laureano-Ortiz, 1992). However, early ANNs remained unstable, susceptible to overfitting, and critically lacked localized spatial awareness (d'Amato and Kauko, 2017).

Spatial Intelligence and Ensemble Methods (2007–2019)

The 2007–2008 financial crisis exposed the devastating consequences of models built on flawed static assumptions, amplifying the need for robustness and locational specificity (d'Amato and Kauko, 2017). This drove the era of Ensemble and Spatial Intelligence (2007–2019). Geographically Weighted Regression (GWR) allowed spatially varying coefficients, improving accuracy by accommodating local market dynamics (d'Amato and Kauko, 2017). The "black box" problem inspired Explainable AI (XAI) techniques like SHAP and Permutation Feature Importance, quantifying feature contributions and introducing transparency (Maselli and Nesticò, 2025). Ensemble methods (Random Forest, XGBoost) aggregated multiple models, minimising prediction errors and reducing overfitting beyond single-network ANNs (Maselli and Nesticò, 2025).

Deep Multimodal Integration (2020–2025)

The final progression into Deep Multimodal and Hybrid AI-BIM Models (2020–2025) arose from realising that property value signals were encoded across heterogeneous formats: images, text, and 3D geometry (Huang et al., 2025). This required specialised processors: Convolutional Neural Networks (CNNs) extract visual features (Poursaeed, Matera and Belongie, 2018), Transformer-based NLP models extract textual nuances (Zhang, 2021), and Graph Neural Networks (GNNs) model spatial dependencies (Zhang, 2021). Sophisticated fusion architectures synthesise these inputs, yielding performance gains of 2.16% to 8.34% over single-modality approaches (Huang *et al.*, 2025).

Legacy of Foundational Concepts and Hybridisation

The journey from econometrics did not involve mere replacement; earlier theoretical concepts remain embedded in modern models. The core insight of hedonic pricing is that value is a function of a bundle of characteristics (Rosen, 1974) still fundamentally dictates feature design and selection in modern AI, guiding which variables (commercial surface, distance to transit) are

integrated into ensemble and deep learning inputs (Maselli and Nesticò, 2025). Similarly, the principles underlying the sales comparison approach and market segmentation underpin strategies for data preprocessing and regional model application, ensuring that model training utilises spatially and structurally homogenous datasets (Demirci, 2021).

This integration is explicitly reflected in instances of hybridisation:

1. **AI-BIM Fusion:** The definitive modern hybridisation involves combining the traditional cost approach with advanced AI. The proposed AI-BIM-driven model uses Building Information Modelling (BIM) to precisely generate 3D data for the deterministic calculation of construction cost, which is then augmented by market-based adjustments derived from Machine Learning (Jafary *et al.*, 2025). This addresses the cost approach's historical flaw of neglecting external market influence (Jafary *et al.*, 2025).
2. **Spatial Econometrics Hybrids:** Techniques like GWR and LVRS are inherently hybrid, augmenting the foundational MRA/HPM framework with sophisticated spatial factors (LAFs or geographically varying coefficients) to overcome simple linearity and spatial blindness (d'Amato and Kauko, 2017).
3. **Non-Deterministic Integration:** Rough Set Theory (RST) and Fuzzy Logic often combine human-derived rules or vague assessments with computational logic, particularly useful in early phases or fragmented markets with limited data, demonstrating that deterministic models are not always the optimal approach (d'Amato and Kauko, 2017).

Reflection on Patterns of Innovation

The history of AVM development exhibits a sequence of disruptive breakthroughs followed by periods of incremental refinement, tightly coupled with the availability of data and computational resources. The shift from MRA to ANNs was fundamentally disruptive, enabled by the dawn of modern computing power, which unlocked non-linear modelling capability (Worzala, Lenk and Silva, 1995).

The subsequent integration of ensemble methods (RF, XGBoost) and spatial models (GWR) was largely incremental but critical, aimed at addressing specific performance issues (overfitting, spatial heterogeneity) exposed when AI was applied to messy, fragmented real estate data (d'Amato and Kauko, 2017). The current phase, defined by deep learning and multimodal fusion, is again disruptive, enabled by the proliferation of Big Data (images, text, GNN nodes) and sophisticated GPU resources required for large-scale training (Huang *et al.*, 2025).

Regulatory and professional acceptance has consistently lagged technological capability. While AVMs are highly efficient (Kok, Koponen and Martínez-Barbosa, 2017), their adoption was hindered by the opacity of the "black box" ANN models (Taffese, 2007). The rise of Explainable AI (XAI) techniques, ensuring that predictions are transparent and justifiable by quantifiable features (like SHAP values), represents the technological response necessary to satisfy regulatory demands for accountability and bolster professional trust (Maselli and Nesticò, 2025). Ultimately, continuous innovation has been necessary not just to enhance accuracy (R^2 moving from ~0.51 in large-scale MRA, Adair and McGreal, 1987, to 0.90-0.95 in multimodal systems, Huang *et al.*, 2025) but to build trust in an industry historically reliant on subjective judgment.

Chapter 5: Literature Map

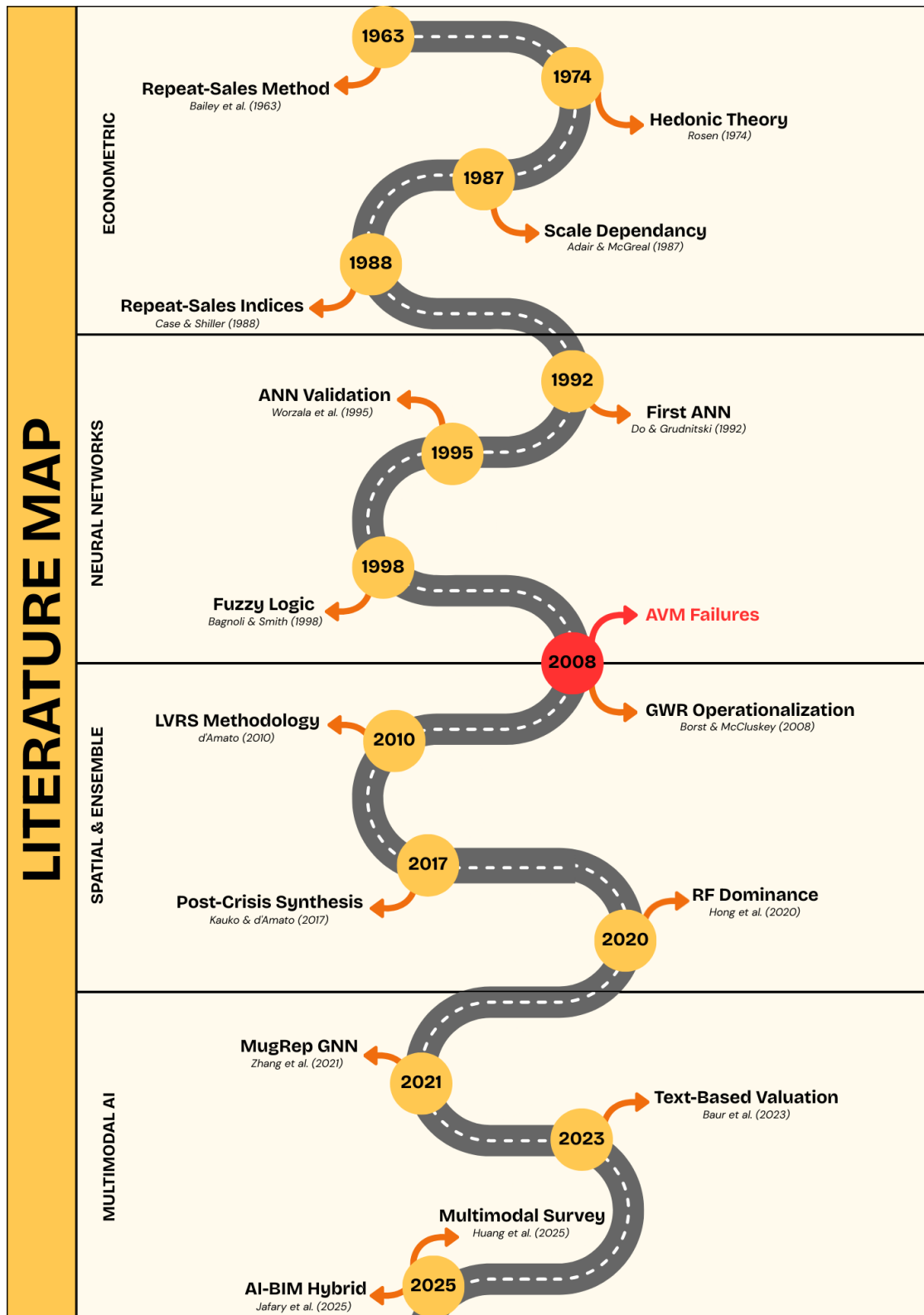


Figure 2: Literature Map

Chapter 6: Significant Contributors

Phase 1: Econometric Foundations

Hedonic Pricing and Regression Models

- Sherwin Rosen (1974) – formalised hedonic price theory, establishing that property prices are implicit equilibrium valuations of bundled attributes; provided the theoretical foundation for all regression-based AVMs
- William C. Pendleton (1965) – early empirical validation of MRA for mass appraisal in the District of Columbia, achieving ~90% variance explanation
- Alastair Adair and Stanley McGreal (1987, 1996) – demonstrated critical scale dependency of hedonic models; showed that micro-market homogeneity dramatically improves accuracy (R^2 from 0.51 to 0.91); normalised MRA within applied valuation practice

Repeat-sales Indices

- Martin J. Bailey, Richard F. Muth and Hugh O. Nourse (1963) – introduced repeat-sales methodology for price index construction, controlling for quality mix by regressing log price changes on time dummies
- Karl E. Case and Robert J. Shiller (1988) – refined repeat-sales with weighted least squares to correct heteroscedasticity; revealed aggregate market predictability but minimal individual property forecasting power

Phase 2: Neural Networks & Non-Linear Modelling (1991–2006)

Artificial Neural Networks

- A.Q. Do and G. Grudnitski (1992) – pioneered ANN application to residential appraisal, demonstrating capacity to model non-linear relationships without explicit functional specification
- Elaine M. Worzala, Margarita Lenk and Ana Silva (1995) – critical validation study; showed ANNs competitive with MRA but warned of architecture sensitivity, inconsistent results across software, and "black box" opacity problems
- Morano (2009, 2015) – applied MLP-ANNs in Italy, achieving high accuracy ($R^2 = 0.99$) and demonstrating robustness to collinearity and heteroscedasticity

Case-based Reasoning and Fuzzy Logic

- González and Laureano-Ortiz (1992) – introduced CBR to real estate, mimicking expert appraiser analogical reasoning through case retrieval and adaptation
- Bagnoli and Smith (1998) – pioneered fuzzy logic for valuation, formalising linguistic uncertainty ("excellent location") via partial set membership

Phase 3: Spatial Intelligence & Ensemble Methods (2007–2019)

Spatial Econometrics and GWR

- Luc Anselin (1988) – foundational work on spatial autocorrelation and heterogeneity; underpins all spatial AVM methods
- Richard A. Borst and William J. McCluskey (2007, 2008) – operationalised GWR and spatial lag models for mass appraisal; demonstrated measurable COD and PRD improvements over OLS
- Farber and Yeates (2006) – showed GWR superior to SAR models in reducing spatial bias

Location Value Response Surfaces

- Maurizio D'Amato (2002, 2004, 2010, 2011) – developed LVRS methodology integrating MRA with continuous location adjustment factors (LAF); introduced valued tolerance relations within rough set theory for rule-based, interpretable valuation in data-scarce markets
- Tom Kauko (2002); Kauko and d'Amato (2011) – extended self-organising maps and rough set methods; co-edited *Advances in Automated Valuation Modelling*, framing post-crisis methodological priorities

Ensemble Learning

- Kok (2017) – demonstrated decision trees outperform HPM for commercial mass appraisal
- Abidoye and Chan (2018) – showed ANNs outperform HPM in Nigeria
- Hong (2020) – RF superior to HPM on 16,601 samples; established ensemble dominance
- Ho (2021); Gao and Shi (2022); Iban (2022) – systematic comparisons confirming XGBoost and RF consistently outperform single models and traditional hedonic methods

Phase 4: Multimodal Deep Learning & AI-BIM Hybrids (2020–2025)

Multimodal Machine Learning

- Chenya Huang, Bin Liang, Zhidong Li and Fang Chen (2025) – comprehensive survey framing real estate as a multimodal ML problem; documented 2–8% accuracy gains from fusing five modalities (attributes, market, text, vision, GIS); first systematic modality classification in the domain
- Poursaeed (2018) – pioneered vision-based valuation using DenseNet for luxury classification from interior images
- Law (2019) – integrated street view and satellite imagery for livability scoring
- Zhang (2024); Baur (2023) – NLP-based valuation; showed text alone achieves $R^2 = 0.79$, with GB+BERT hybrid yielding lowest errors

Graph Neural Networks and Spatio-temporal Models

- Fu (2014, 2016) – emphasised geographic dependence; integrated POI, transport and mobility data
- Zhang (2021) – MugRep hierarchical graph framework (MAE = 0.32 vs LR = 0.48)
- Lee (2023) – ST-RAP spatio-temporal GCN+LSTM (MAE = 21.77 vs LR = 105.09)

AI-BIM Integration

- Peyman Jafary, Davood Shojaei, Abbas Rajabifard (2025) – first end-to-end hybrid AI-BIM framework combining ML and AVMs, BIM QTO, NLP cost matching, FM-based depreciation and ML market adjustment; demonstrated 100% accuracy within transaction range for a Melbourne high-rise
- Arcuri and Ruggiero (2020); Khodabakhshian and Toosi (2021) – earlier proof-of-concept BIM-cost integrations, but static and lacking market linkage

Explainable AI

- Maselli and Nesticò (2025) – applied SHAP and permutation importance to RF and XGBoost, addressing "black box" transparency for professional acceptance

Industry and Standards Bodies

- International Association of Assessing Officers (2003) – first formal AVM standard; defined model specification, calibration, testing (COD, PRD)
- Royal Institution of Chartered Surveyors (2013) – European AVM information paper; formalised hybrid valuation (AAAVM, SAAVM, AVMAA) and confidence-level requirements
- Rating agencies (Fitch 2007; S&P, Moody's) – operational AVM criteria for securitisation and lending; bridged academic models and commercial deployment

Chapter 7: Discussion and Synthesis

Property valuation technology has undergone a foundational four-phase transformation, shifting from subjective heuristics to integrated, data-driven intelligence. This evolution began with the econometric foundation (Phase 1), where Multiple Regression Analysis (MRA), despite introducing statistical objectivity, proved fundamentally inadequate for capturing market non-linearity and geographical heterogeneity (Rosen, 1974; Adair and McGREAL, 1988). This technical limitation drove Phase 2's pursuit of non-linear algorithms, primarily Artificial Neural Networks (ANNs), which offered superior pattern recognition without rigid functional forms but suffered from instability, inconsistency, and the "black box" problem (A. Quang Do, 1992; Worzala, Lenk and Silva, 1995).

Technical Evolution and Spatial Intelligence

The post-2008 financial crisis accelerated development into robust ensemble learning models of Phase 3, epitomised by Random Forest and XGBoost. These algorithms addressed earlier AI instability, providing models robust against noise and heterogeneity, drastically improving practical accuracy and scalability in Automated Valuation Models (AVMs) (Maselli and Nesticò, 2025; Stojanović, Galić and Kahrić, 2025). Concurrently, sophisticated spatial intelligence through Geographically Weighted Regression (GWR) and Location Value Response Surface (LVRS) models, allowed spatially varying coefficients, resolving traditional MRAs' localised market failures (d'Amato and Kauko, 2017).

Phase 4's culmination focuses on deep learning and multimodal fusion. Current models process diverse data modalities, integrating structured attributes with previously inaccessible vision (Convolutional Neural Networks), textual descriptions (Transformer models), and geospatial data (Graph Neural Networks) into unified frameworks (Zhang, 2021; Huang *et al.*, 2025). Multimodal integration fundamentally enhances accuracy, with performance improvements of 2.16% to 8.34% over single-modality approaches, achieving R^2 scores up to 0.95 (Ali *et al.*, 2025; Huang *et al.*, 2025). Hybrid AI-BIM frameworks efficiently combine cost-based data from 3D models with market adjustments, demonstrating prediction errors as low as 0.057% (Jafary *et al.*, 2025).

Addressing Original Challenges and Persistent Limitations

Contemporary approaches have substantially addressed Phase 1 challenges. Flexible ensemble and deep learning models process vast, multi-dimensional datasets, overcoming linear models' unsuitability for property heterogeneity (Zhang, 2021). AI-driven inputs minimise human bias and enhance reproducibility (Shetty *et al.*, 2020; Maselli and Nesticò, 2025). However, models still struggle with unexplained variance from unobservable micro-externalities and lack the interpretability necessary for widespread acceptance (d'Amato and Kauko, 2017; Maselli and Nesticò, 2025).

Social Impact and Interpretability Trade-offs

The fragility of pre-2008 methods demonstrated that inaccurate valuation threatens financial stability (d'Amato and Kauko, 2017). Modern AVMs support regulatory compliance for loan risk assessment and democratize valuation, enabling non-experts to access reliable estimates through user-friendly applications (Stojanović, Galić and Kahrić, 2025). Regulatory frameworks like the EU AI Act enforce transparency requirements for high-risk real estate AI systems (Yagmur, 2025), shifting workforce demands toward oversight and expert interpretation rather than subjective judgment (Lausberg and Dust, 2017).

The interpretability-performance trade-off persists critically. Complex models achieve superior accuracy but operate as "black boxes" (Ali *et al.*, 2025). Explainable AI (XAI) techniques like SHAP and Permutation Feature Importance address this by quantifying feature contributions, ensuring transparent, justifiable valuations (Maselli and Nesticò, 2025). Hybrid AI-BIM models balance this trade-off by grounding predictions in verifiable cost-based inputs while augmenting with market-driven adjustments (Jafary *et al.*, 2025).

Future Directions

Research gaps remain clear. Future efforts must develop fair, robust models across global markets with data variability, integrate real-time streaming data for dynamic valuations, and incorporate richer micro-externalities to explain residual variance (d'Amato and Kauko, 2017; Maselli and Nesticò, 2025; Yagmur, 2025). Technological adoption must accelerate through Large Language Model integration to fully capitalise on nuanced property listing data (Huang *et al.*, 2025).

Implementations

Historical Approach: Hedonic Regression (Phase 1)

The hedonic pricing model, pioneered by economist Sherwin Rosen (1974), represents the foundational approach to automated property valuation, decomposing house prices into implicit values of individual property characteristics. The implementation utilises Ordinary Least Squares (OLS) linear regression on the Boston Housing dataset (506 observations, 13 features) to predict property prices based on attributes including crime rate (CRIM), average number of rooms (RM), property tax rate (TAX), nitric oxide concentration (NOX), and neighbourhood demographics.

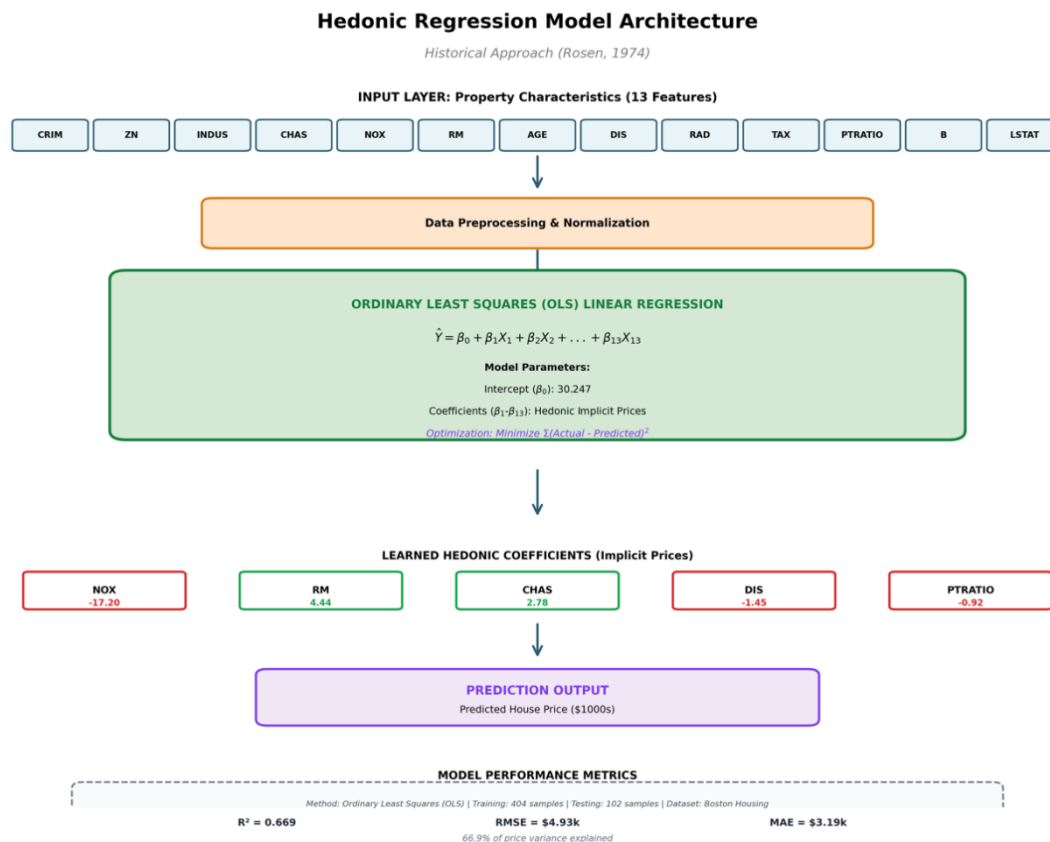


Figure 3: Hedonic Regression Model Architecture

The model architecture (Figure 3) follows a classical pipeline: raw property characteristics are preprocessed and normalised before being fed into the OLS regression engine, which optimises coefficients (β_0 - β_{13}) by minimising the sum of squared residuals through the closed-form solution $\hat{\beta} = (X'X)^{-1}X'y$.

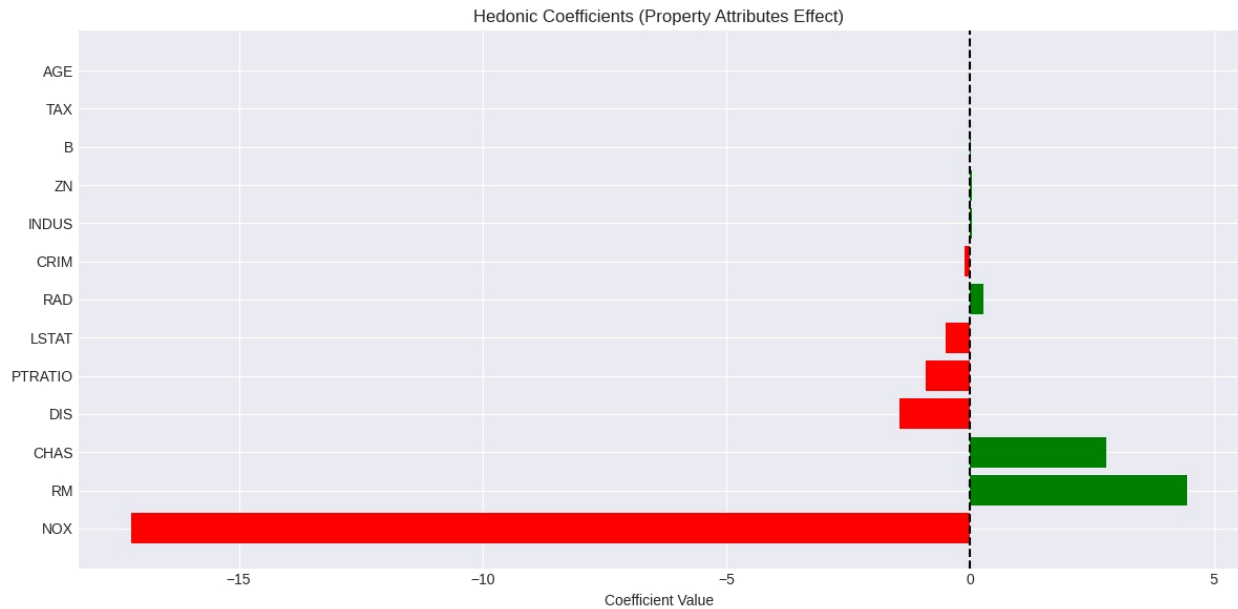


Figure 4: Hedonic Coefficients

The learned hedonic coefficients (Figure 4) reveal interpretable implicit prices, with NOX (air pollution) showing the strongest negative impact (-\$17,203 per unit increase), while RM (average rooms) demonstrates the most positive effect (+\$4,439 per additional room), quantifying market preferences that were previously only understood intuitively by appraisers. The model achieved moderate performance with $R^2 = 0.669$ (explaining 66.9% of price variance), RMSE of \$4,929, and MAE of \$3,189 on the test set.



Figure 5: Actual vs Predicted Prices

(Figure 5), demonstrating the interpretability and transparency of linear hedonic models while simultaneously exposing their fundamental limitation, the assumption of purely linear relationships between property attributes and market value fails to capture complex non-linear interactions and spatial dependencies. Despite these constraints, the hedonic regression framework established the theoretical foundation for all subsequent property valuation methodologies, introducing the critical concept of decomposing heterogeneous goods into measurable, valued components that could be systematically analysed through statistical methods.

Intermediate Approach: XGBoost Ensemble Learning (Phase 3)

XGBoost (Extreme Gradient Boosting) represents a significant algorithmic advancement in property valuation, transitioning from simple linear relationships to complex ensemble learning that captures non-linear feature interactions and hierarchical patterns in real estate data (Sharma et al., 2024). The implementation utilises the Kaggle House Prices dataset comprising 1,460 training samples with 81 features, including physical attributes (OverallQual, GrLivArea, GarageCars), temporal characteristics (YearBuilt, YearRemodAdd), and neighbourhood indicators, which undergo comprehensive preprocessing including missing value imputation, categorical encoding via LabelEncoder, and outlier removal using the interquartile range (IQR) method, reducing the dataset to 1,399 samples.

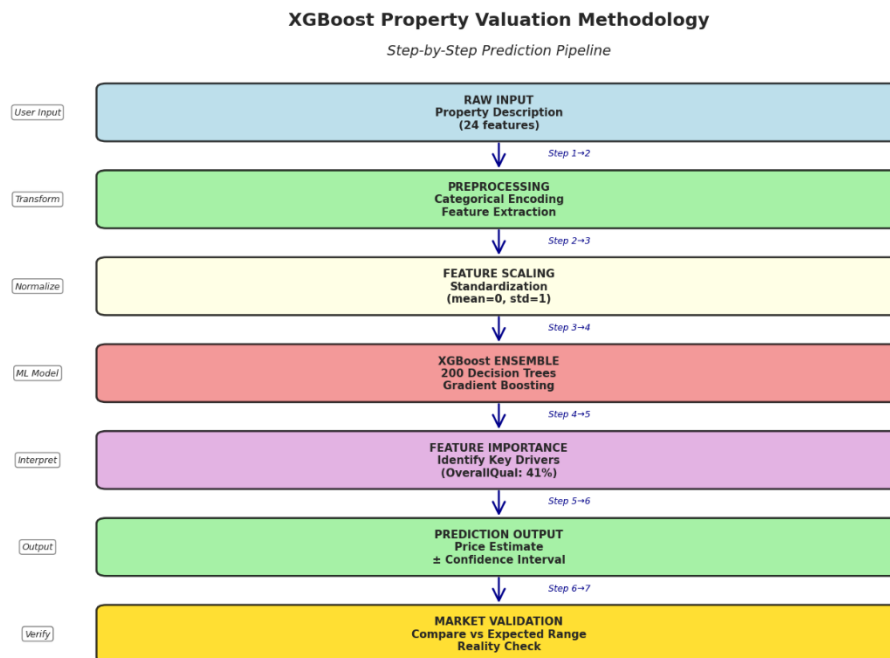


Figure 6: XGBoost Model Architecture

The XGBoost architecture (Figure 6) employs gradient boosting methodology where an ensemble of 200 shallow decision trees ($\text{max_depth}=5$) is trained sequentially, with each subsequent tree learning to correct residual errors from previous trees through additive modelling: $\hat{y} = \sum f_k(x)$ for $k=1$ to 200, where f_k represents individual tree predictions weighted by $\text{learning_rate}=0.05$ to prevent overfitting. Hyperparameter optimisation through GridSearchCV explored 729 parameter combinations across max_depth (5,7,9), learning_rate (0.05-0.15), $n_estimators$ (100-300), and regularisation terms ($\text{subsample}=0.8$, $\text{colsample_bytree}=0.8$), converging on an optimal configuration that balances model complexity and generalisation capacity.

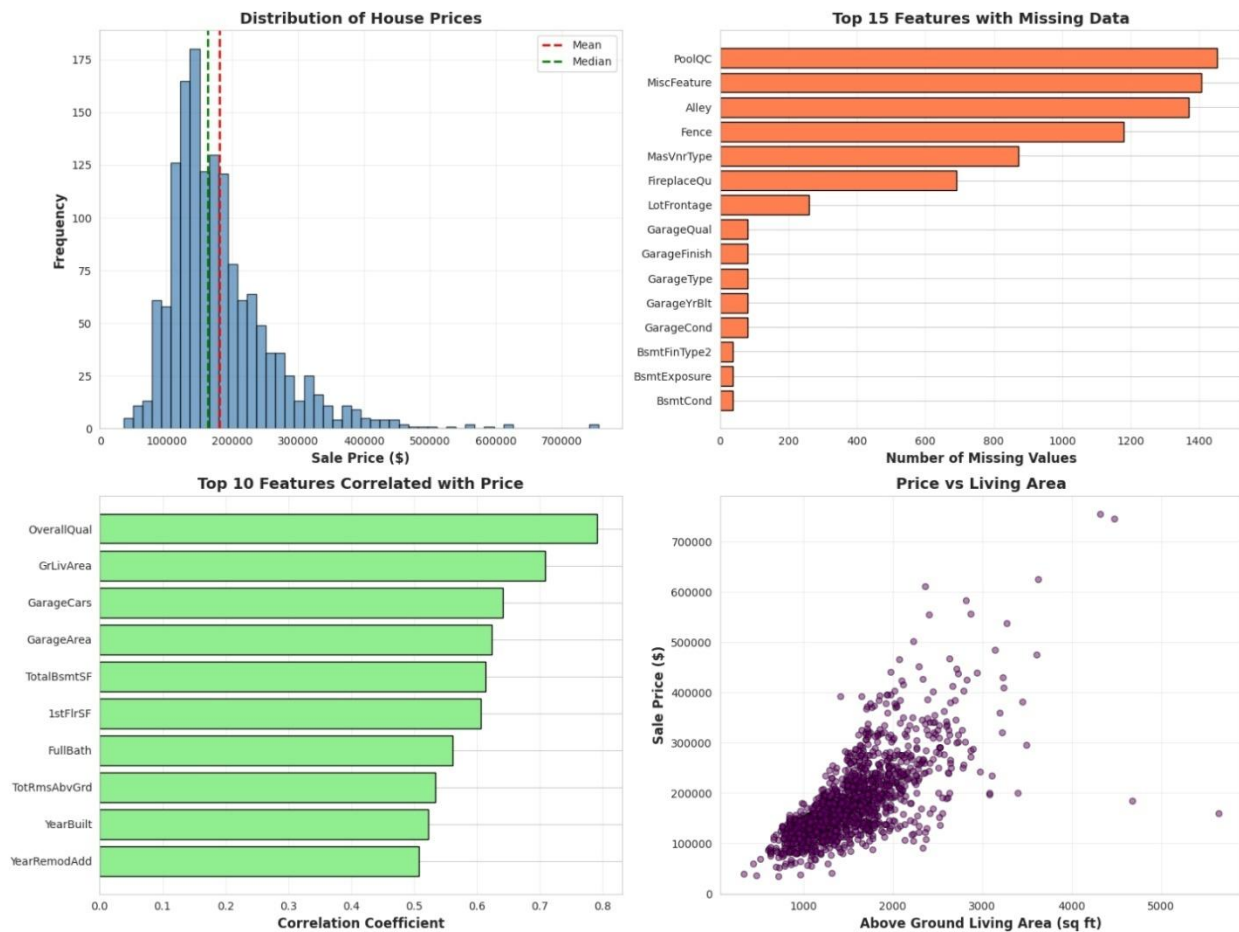


Figure 7: Exploratory Data Analysis

Exploratory data analysis (Figure 7) revealed right-skewed price distributions with significant missing data in features like PoolQC and MiscFeature, while correlation analysis identified OverallQual (0.79), GrLivArea (0.71), and GarageCars (0.64) as strongly associated with sale prices.

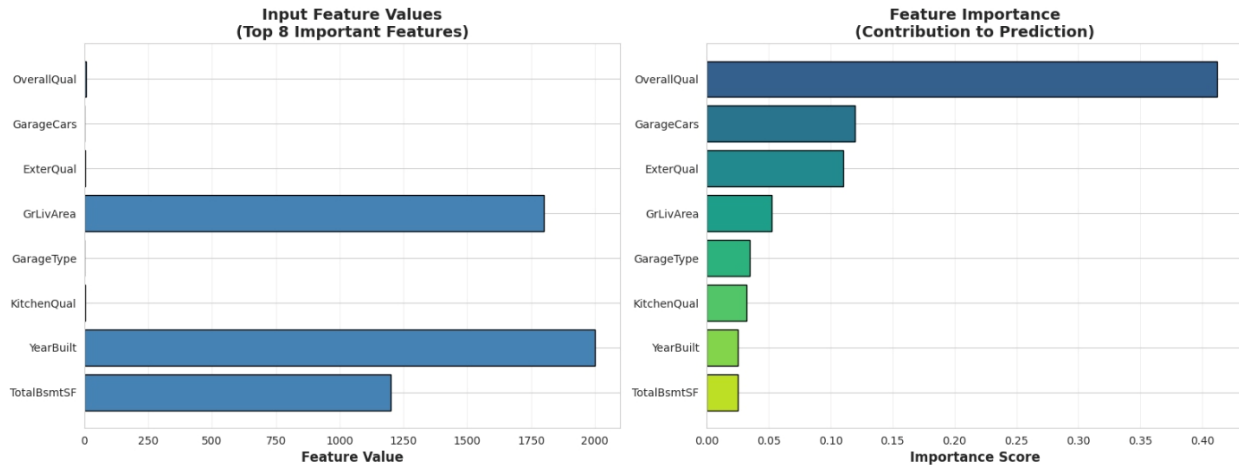


Figure 8: Feature Importance Analysis

Feature importance analysis (Figure 8) reveals OverallQual as the dominant predictor (41.23% importance), followed by GarageCars (11.94%) and ExterQual (10.99%), demonstrating XGBoost's ability to automatically identify non-linear value drivers without manual feature engineering, a stark contrast to the predetermined linear coefficients in hedonic regression.

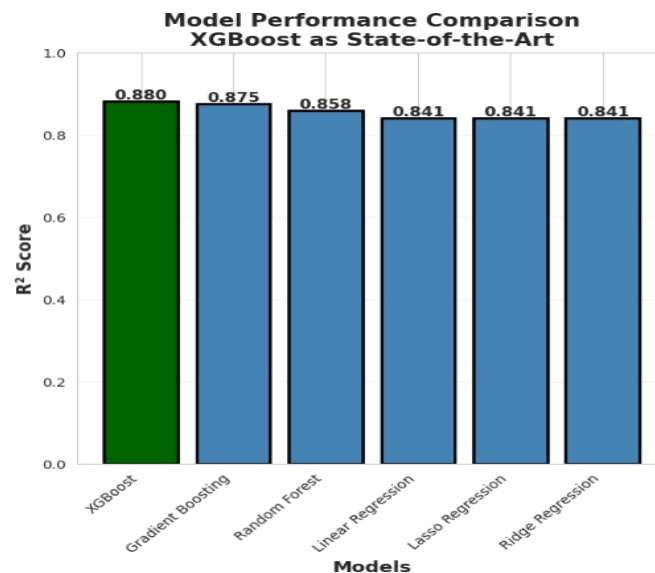


Figure 9: Model Performance Comparison

The tuned model achieved superior performance with $R^2=0.8824$ (explaining 88.24% of price variance), $RMSE=\$19,342$, and $MAE=\$13,834$, representing a 31.6% improvement in explained variance over the historical hedonic approach ($R^2=0.669$), validated through 5-fold cross-validation yielding 87.91% consistency.

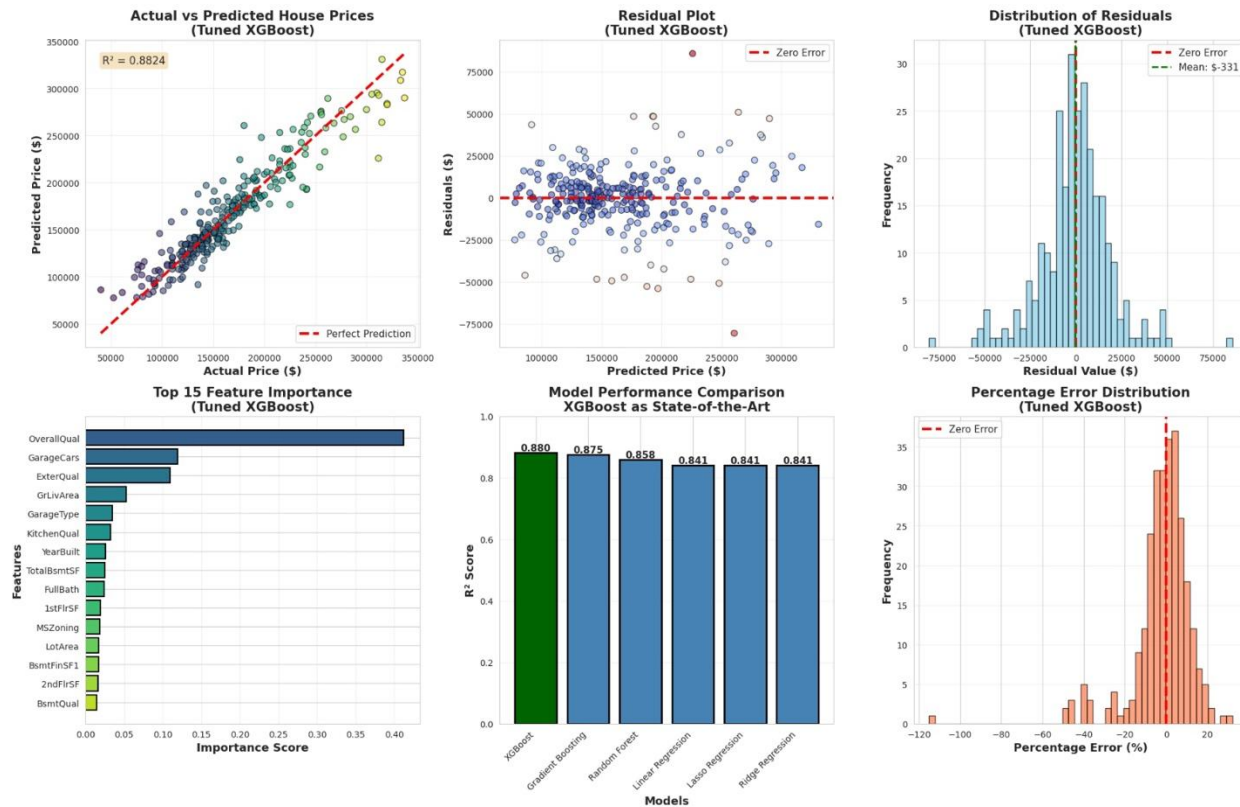


Figure 10: Comprehensive Model Validation

The prediction methodology (Figure 10) demonstrates sophisticated error correction: raw property features are standardised (mean=0, std=1), then propagated through the ensemble, where each tree votes on price adjustments, with the final prediction aggregating 200 tree outputs while maintaining interpretability through feature importance values that quantify individual feature contributions. Unlike linear models that assume constant marginal effects, XGBoost captures threshold effects (e.g., quality ratings above 8 exponentially increase value) and interaction patterns (e.g., large living area combined with excellent condition multiplies value beyond additive effects), revealed through decision tree splits that partition the feature space into heterogeneous pricing regions. The model's robustness is evidenced by consistent performance across market segments, with prediction confidence intervals (68% CI: $\pm\$19,342$) quantifying uncertainty and enabling risk-adjusted valuation decisions for real estate stakeholders. This ensemble approach establishes XGBoost as the contemporary standard for automated property valuation, bridging the gap between classical econometric interpretability and modern machine learning predictive power, while setting the foundation for deep learning architectures that further exploit spatial dependencies and unstructured data sources.

Recent Approach: Multimodal Embedding Fusion with Advanced Ensemble Models (Phase 4)

Our multimodal property price prediction system was implemented following the methodology proposed by Hasan et al. (2024), with significant enhancements and optimisations. The pipeline consists of five sequential steps:

1. Data Preprocessing – we processed 53,220 property records from RightMove, cleaning textual descriptions, extracting structured features, and organising image URLs for multimodal learning
2. Graph Embeddings – we constructed a property-locality bipartite graph connecting properties to their localities and applied Node2Vec with optimised hyperparameters (walk length: 80, context size: 10, embedding dimension: 64) to capture spatial relationships and neighbourhood characteristics
3. Text Embeddings – property descriptions were encoded using Sentence-BERT (all-mpnet-base-v2 model) to generate 768-dimensional semantic embeddings capturing textual nuances about amenities, conditions, and unique features
4. Image Embeddings – property images were processed through CLIP (ViT-B/32 vision encoder) to extract 512-dimensional visual representations encoding architectural style, property condition, and aesthetic appeal
5. Multimodal Fusion & Model Training – all embeddings were concatenated (total: 1,344 features) and evaluated using two approaches:
 - I. The author's approach replicated the original methodology using 7 regression models (Lasso, ElasticNet, Gradient Boosting, XGBoost, LightGBM, Averaging Ensemble, and Stacking Ensemble) with log-normalisation and 99.9th percentile outlier removal, achieving best performance with the Stacking Ensemble (Test RMSE: \$197,965, R^2 : 0.7356, 43.9% predictions within 10% error).
 - II. Our custom approach introduced 6 models, including a Multimodal Fusion, Simple MLP, XGBoost, LightGBM, Averaging Ensemble and Weighted Ensemble. Our best model, the Multimodal Network, outperformed the author's approach by 15.1% (Test RMSE: \$168,063, R^2 : 0.8094, 49.8% within 10% error).



Multimodal Property Price Prediction Architecture

Based on Hasan et al. (2024) with Custom Enhancements

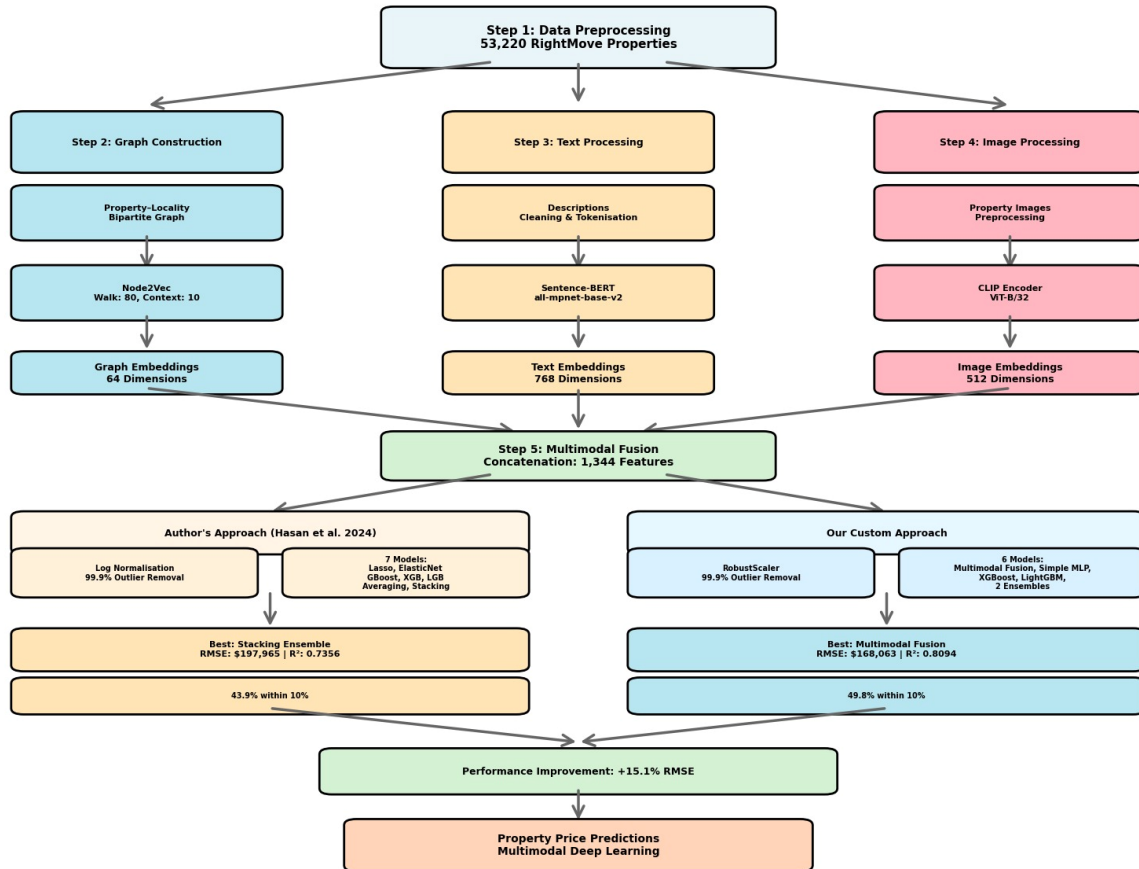


Figure 11: Multimodal System Architecture

Chapter 8: Conclusion

The review investigates the technological evolution of Automated Valuation Models (AVMs) and Artificial Intelligence (AI) integration in property valuation, aiming to enhance predictive accuracy, consistency, and objectivity beyond subjective conventional methods.

The evolution began with the Econometric Foundation (Phase 1, 1963–1991), centred on Multiple Regression Analysis (MRA) and the Hedonic Price Model (HPM) (Rosen, 1974). This established scientific valuation but was limited by linearity assumptions, performing poorly outside homogeneous markets (Adair and McGREAL, 1988). Phase 2 (Neural Networks and Non-Linear Modelling, 1991–2006) utilised Artificial Neural Networks (ANNs) to overcome linear constraints and Fuzzy Logic for qualitative data (Worzala, Lenk and Silva, 1995; Bagnoli and Smith, 1998). Phase 3 (Ensemble Methods and Spatial Intelligence, 2007–2019), spurred by the financial crisis, prioritised robustness and location-awareness through Geographically Weighted Regression (GWR), Random Forest (RF), and XGBoost. Phase 4 (Deep Learning and Multimodal Fusion, 2020–2025) represents the greatest leap, synthesising heterogeneous sources; vision (CNNs), text (Transformers), structured data, and geospatial information significantly boosting accuracy (Huang *et al.*, 2025).

Location intelligence proves critical; predictive accuracy hinges on modelling localised submarket dynamics through GWR and Location Value Response Surface (LVRS) models. Multimodal integration in Phase 4 demonstrates substantial performance gains over single-modality approaches. However, persistent challenges include data-quality risks contributing to overfitting (Maselli and Nesticò, 2025) and the "black box" nature of potent AI models, conflicting with transparency requirements, necessitating Explainable AI (XAI) techniques (Maselli and Nesticò, 2025).

Researchers must advance XAI techniques (SHAP, PFI) to resolve interpretability-accuracy trade-offs. Practitioners: valuers, banks, regulators, benefit from objective AVM outputs (Stojanović, Galić and Kahrić, 2025) but require transparency for professional justification. System designers must implement robust ensemble algorithms and multimodal fusion architectures.

Future trajectories involve integrating advanced Large Language Models (LLMs) and deep neural networks while enhancing transparency through XAI and addressing ethical frameworks to ensure trust in automated decision-making processes.

GitHub Repository

[Access the GitHub repository here.](#)

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